

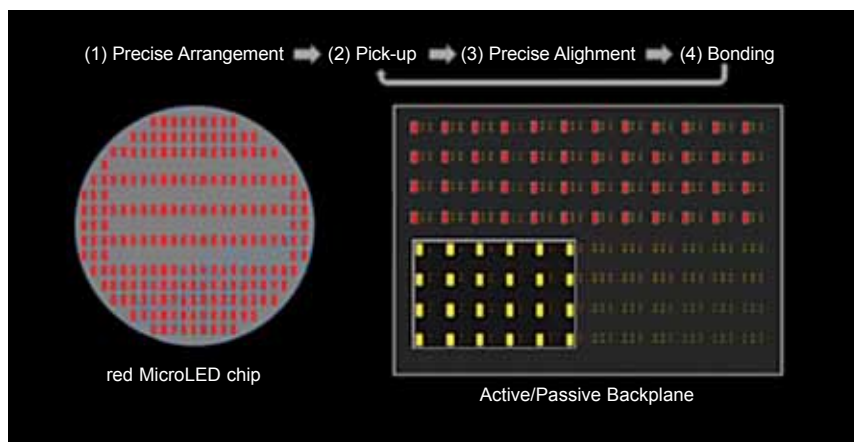


Fig. 3: Mass transfer for μ LED chips (Source: PlayNitride)

tion. The panels operated in bright rooms need a high luminant display, and μ LEDs have proven to be the most efficient followed by OLEDs. TCL, a Chinese electronics firm, recently launched its new μ LED-powered smart glasses.

μ LEDs have also become a popular embellishment on functional applications like clothing or protective jackets for emergency service personnel. Their modular design of multiple same-size display panels teaming up to make a large-format screen, and their creative aspects of flexibility, high transparency, and ability to attach with sensors, make them a top choice for manufacturers.

For instance, a conventional mobile screen would not easily integrate with non-display components like touch, however a μ LED can integrate many functional devices due to the considerable gap between each pixel inside the panel.

The big hurdle: Manufacturing

In spite of the huge number of applications, the process of manufacturing μ LEDs is a lot more difficult than it sounds. μ LED is simply an LED that converts electrical energy into light. But the extremely small μ LEDs face numerous challenges in manufacturing as they require specific chip designs, manufacturing infrastructure, and assembly. Whether at wafer stage, post deposition, or at

the assembling level, μ LED makers need to have a reliable inspection.

“Traditionally the (LED) manufacturing process includes epitaxy and chip process, and it is the same for μ LED. But higher performance is needed for the chip process. Mass transfer is another stage where we have no special equipment in the market,” says Falcon Liu, marketing director, PlayNitride, a manufacturing company in Taiwan.

First of all, the μ LED epitaxy needs to be improved in terms of both wavelength and uniformity in thickness. The uniformity makes the wavelength more concentrated and thus can reduce the backend fine-tuning cost. But if the size is reduced to less than 100 micrometres, there can be power leakages due to non-uniformity while cutting around the chip.

The LED technology is not new but manufacturing a panel array with tiny components makes it challenging. μ LEDs are microscopic

μ LED manufacturing challenges

- Equipment accuracy
- Transfer yield
- Time of mass transfer process
- Manufacturing technology
- Inspection method
- Rework
- Processing cost

versions of an LED without a package. Imagine making a μ LED TV panel. It may roughly require about 24 million individual μ LEDs, which means manufacturing 24 million μ LEDs and then transferring them to backplane of the TV. This mass transfer is the unresolved bottleneck for panel manufacturers.

One potential solution is the pick and place technique, even if it does not meet the $\pm 1.5\mu\text{m}$ accuracy requirement. Flip chip technique is another one that is being widely followed, where a wafer carrying the LED layer is flip-bonded onto the driver circuit and soldered. This method is, however, not quite feasible as only one substrate is flipped at a time, and there are high chances for pesky alignments and thermal mismatch.

Liu points out there are many ways of manufacturing for each application. He adds, “Production equipment is ready in the market, but needs modification. We need to find the correct process to improve yield and manufacturing time. Most of the equipment innovations are around mass transfer techniques. PlayNitride follows a stamping process, like pick up and place. “We produce Epi wafers by ourselves through a stamp to transfer μ LED chips from wafer to backplane.”

“I think LED manufacturing is at a matured stage, but the manufacture of microchips needs careful understanding and effort. For μ LEDs, the key is to make the manufacturing process rapid and massive. We need mass inspection and testing. A 4K display may need roughly 24 million μ LEDs. It is not feasible to pick one by one,” says Liu.

He observes that manufacturers are concentrating on mass transferring to improve yield. “We are thinking on how to make it more efficient and perfect for mass production. We try to find new solutions, if it does not work, we keep innovating. Pick and place method is relatively